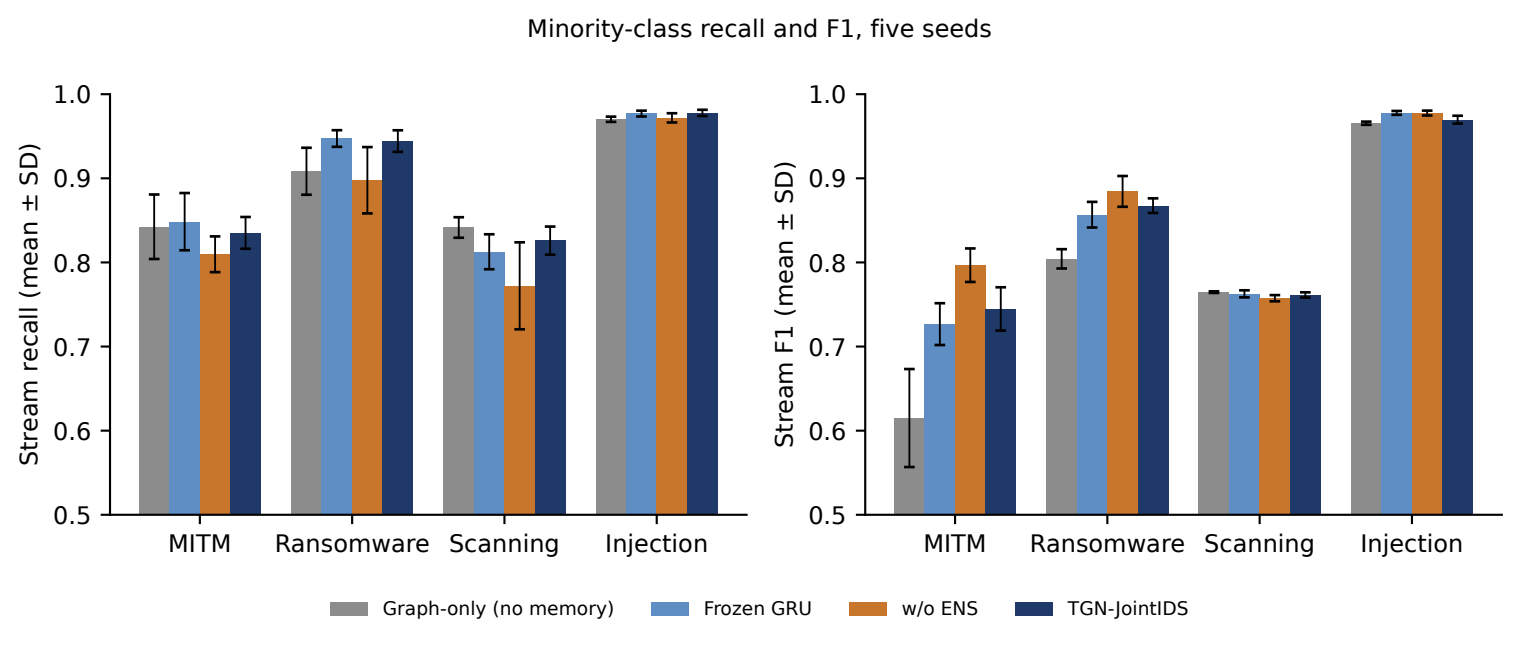
Minority-class recall and F1, five seeds
Stream recall (mean ± SD)
Stream F1 (mean ± SD)
1.0
0.9
0.8
0.7
0.6
0.5
MITM
Ransomware
Scanning
Injection
Graph-only (no memory)
Frozen GRU
w/o ENS
TGN-JointIDS